

How Much Target Data Is Needed?

Sample-Efficient Adaptation of 12-Lead ECG Classifiers Under Distribution Shift

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Abstract

Deep ECG classifiers trained on one clinical dataset are commonly assumed to perform worse on a different population, but the size, significance, and clinical meaning of such a change is rarely quantified for a specific model and dataset triplet. PTB-XL (21,837 records), MIMIC-IV-ECG (~800,000 records), and Chapman-Shaoxing (10,646 records) are independent public 12-lead ECG datasets enabling a controlled cross-dataset study [3–5]. This project trains a 1D-ResNet on each dataset in turn and evaluates it out-of-distribution (OOD) on the other two (a three-way rotation), quantifying the size and significance of any cross-dataset change, including whether OOD performance stays distinguishable from a random classifier. It then studies how much labelled target-domain data expressed as a *percentage* of the target adaptation pool, $P \in \{5, 10, 20, 30, 50\}\%$, rather than a fixed count is needed before two distinct classes of adaptation, post-hoc calibration (Temperature Scaling) and parameter adaptation (Linear Probing, LoRA), yield only marginal further improvement. Rather than asking whether OOD performance “recovers” to a source value that may be unattainable when the target population genuinely differs, the project defines sample-dependent stabilization/improvement criteria, reported with bootstrap uncertainty, as a percentage-based audit framework for adapting an imported ECG model to a new population.

I. Research Problem and Background

External validation of a model on data separate from its training population, with sufficiently precise estimates, is standard practice [1, 2]. Whether a 12-lead ECG classifier trained on one public dataset performs worse on another is an empirical question rather than a given: BenchMD’s PTB-XL→Chapman-Shaoxing OOD track [9], Strodthoff *et al.*’s transfer to ICBE2018 [8], and HeartBEiT’s low-sample comparisons [10] give indirect evidence that cross-dataset ECG performance can change, but none reports, for a fixed backbone and this PTB-XL/MIMIC-IV-ECG/Chapman-Shaoxing triplet, the size of that change, its significance, or whether a shifted model still carries transferable signal or is reduced to a random classifier nor traces a percentage-indexed stabilization curve across adaptation strategies with a rotated source and an uncertainty-quantified diminishing-return threshold. Both gaps are addressed directly here.

A further issue motivates the design: suppose source in-distribution AUROC of 0.93 falls to 0.78 OOD and rises to only 0.84 after adaptation on a large target sample. Concluding the model “failed to recover” would wrongly imply insufficient target data, when the target may simply differ in label definitions, prevalence, population, or acquisition hardware [3–5]. The source value is therefore never treated as an attainable target; instead the project asks how OOD performance *stabilizes and improves* with more adaptation data, reporting the gap to source only as descriptive context.

the OOD adaptation pool, is needed before calibration and parameter-adaptation strategies yield only marginal improvement in a PTB-XL/MIMIC-IV-ECG/Chapman-Shaoxing rotation, and how large and significant is the underlying cross-dataset change to begin with?

II. Proposed Solution

Each dataset (Table 1) serves in turn as *source*: a 1D-ResNet is trained on it and evaluated OOD, unadapted, on the remaining two *targets*, giving six source→target pairs and a baseline M_0 per pair, tested for significance against chance (AUROC= 0.5) and reported not judged against the source in-distribution value.

For each pair, a held-out target evaluation set is fixed and the remainder forms an adaptation pool, sampled at

$$P \in \{5, 10, 20, 30, 50\}\% \quad (1)$$

of that pool rather than a fixed count, keeping results comparable across datasets differing by orders of magnitude (absolute N is reported alongside P for transparency).

Two classes of adaptation are compared: **post-hoc calibration** via Temperature Scaling, which reshapes probabilities without changing ranking and so cannot change AUROC [6]; and **parameter adaptation** via Linear Probing and LoRA, which retrain part of the model and can change both discrimination and calibration [7]. The two classes are scored on different outcome sets accordingly (Section VI), across ≥ 30 Monte Carlo repetitions per (pair, method, P), increased where confidence intervals are unstable.

III. Research Objectives

A. Quantify Cross-Dataset Performance Change

Measure the size and significance of the unadapted OOD change per pair, and whether OOD AUROC stays significantly above chance.

B. Estimate Percentage-Dependent Stabilization/Improvement

Measure how AUROC, Brier score, and ECE change with adaptation percentage, relative to each pair’s own unadapted baseline.

C. Compare the Two Classes of Adaptation

Compare calibration vs. parameter adaptation on stabilization, improvement, variability, and cost, respecting that only the latter can change discrimination.

D. Determine Diminishing-Return Thresholds

Estimate rotation/dataset/method-specific $P_{\min}$ with uncertainty, under predefined criteria, not as a universal requirement.

E. Develop a Percentage-Based Audit Framework

Package the above into a reproducible procedure for auditing an imported ECG model against a new population.

IV. Related Work

External validation. Collins *et al.* show validation size affects estimate precision [1]; Riley *et al.* give a model/outcome/setting-

How much labelled target-domain data, as a percentage of

specific sample-size framework [2], supporting a condition-specific $P_{\min}$ over one universal number.

Cross-dataset ECG evidence. BenchMD’s PTB-XL→Chapman OOD track [9] is the closest existing evidence for the claim under study; Strodthoff *et al.* [8] and HeartBEiT [10] add transfer and low-sample evidence. This project extends them with significance testing, MIMIC-IV-ECG, a rotated source, and adaptation-strategy comparison.

Calibration and parameter-efficient adaptation. Guo *et al.* show temperature scaling is a simple, effective post-hoc calibrator that cannot alter ranking [6], motivating its own outcome set here. LoRA [7], developed for language models, is tested for ECG classification as an open question, including whether it moves discrimination as well as calibration.

Sampling grid. Few-shot ECG studies typically use absolute, study-specific N grids (e.g. Markov *et al.* [11]); because the three target pools here differ by orders of magnitude, a percentage-of-pool grid is used instead for cross-dataset comparability.

V. Methodology

Fig. 1 summarizes the workflow: each dataset trains as source and is evaluated OOD on the other two; calibration and parameter adaptation are then compared across percentage levels to estimate $P_{\min}$ per rotation.

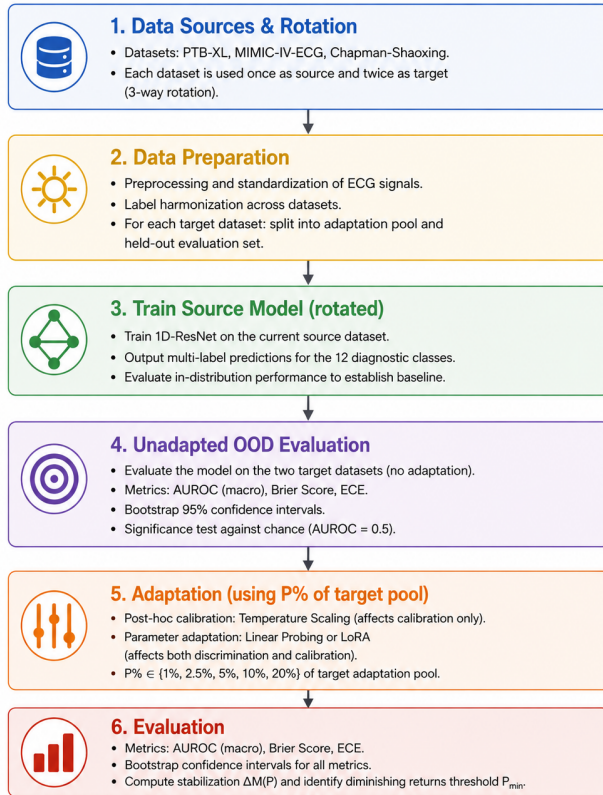


Figure 1: Three-way source/target rotation, unadapted OOD evaluation, and percentage-based stabilization/improvement audit ($P_{\min}$) across calibration and parameter adaptation.

A. Datasets and Rotation Design

Each dataset serves once as source and twice as target (six pairs total), with a common input representation and documented label harmonization; unmapped labels are not treated as equivalent. Each source model is trained with recorded preprocessing, optimizer,

Table 1: Datasets

Dataset	Role	Size / Evidence
PTB-XL	Source/Target	21,837 records/18,885 pts, expert-annotated [3]
MIMIC-IV-ECG	Source/Target	~800k records/~160k pts, 500 Hz, credentialed
Chapman-Shaoxing	Source/Target	PhysioNet access [4]
		10,646 records, 500 Hz, rhythm labels [5]

schedule, and seeds. Table 2 gives the published PTB-XL in-distribution range, used only as descriptive context, never as a recovery target for OOD results.

Table 2: Published PTB-XL Reference Macro-AUROC (In-Distribution Context)

Task	Macro-AUROC	Source
Diagnostic superclass	≈0.92–0.93	[8, 12]
Form statements	≈0.83–0.89	[8, 12]
Rhythm statements	≈0.93–0.96	[8, 12]

B. Adaptation Methods and Pipeline

Post-hoc calibration fits a scalar T : $p_i = \text{softmax}(z_i/T)$, $T > 0$, on frozen logits, changing probabilities but not ranking [6]. Parameter adaptation retrains only the final head (Linear Probing) or trains low-rank updates $W' = W + BA$ with W frozen (LoRA) [7], measuring trainable-parameter count rather than assuming it. Pipeline per pair: source training → unadapted OOD evaluation (with a chance-level significance test) → percentage sampling → adaptation → independent held-out evaluation, $k \geq 30$ Monte Carlo repetitions per (pair, method, P). Stabilization trajectories are initially summarized by $y(P) = aP^{-b} + c$, with diagnostics determining adequacy and alternative fits used otherwise.

VI. Evaluation Metrics and Audit Thresholds

AUROC (macro-averaged), Brier score, and ECE (ECE = $\sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|$) are reported for parameter adaptation; Temperature Scaling’s outcome variables are Brier score and ECE only, with AUROC as a manipulation check expected to stay unchanged. For ordered percentages $P_1 < \dots < P_5$, metric M_P , and unadapted baseline M_0 , improvement relative to the pair’s own baseline (not the source value) is

$$\Delta M(P_k) = s (M_{P_k} - M_0), \quad s = \begin{cases} +1 & \text{AUROC} \\ -1 & \text{Brier, ECE} \end{cases} \quad (2)$$

and the diminishing-return threshold is

$$P_{\min}^{(\Delta)} = \min \{P_k : |M_{P_{k+1}} - M_{P_k}| < \tau\} \quad (3)$$

with τ predefined per metric (0.01 for AUROC/Brier, 0.005 for ECE). The gap to source, $G(P_k) = M_{P_k} - M_{\text{source}}$, is reported descriptively alongside ΔM and $P_{\min}^{(\Delta)}$, but closing it is never a pass/fail criterion, since a persistent gap may reflect genuine target-population differences rather than insufficient data.

VII. Controls, Reproducibility, and Limitations

The same protocol applies across all six pairs; adaptation and evaluation samples are disjoint; seeds are recorded. The repository will include preprocessing, label-mapping, model, pipeline, and analysis code, configs, results, and documentation, without redistributing restricted patient-level data [4].

Limitations: (i) dataset differences (institution, population, acquisition) are reported as cross-dataset shifts, not attributed to geography alone [3–5]; (ii) $P_{\min}$ is rotation/dataset/method-specific, not universal [2]; (iii) rotating the source triples the training/evaluation matrix relative to a single-source design, so $k \geq 30$ is conservative and raised only where compute allows; (iv) $P = 5, 10\%$ may be unstable at small absolute N (e.g. Chapman-Shaoxing), mitigated via Monte Carlo repetition and bootstrap CIs; (v) Temperature Scaling’s AUROC is fixed by construction and so is compared to Linear Probing/LoRA only on calibration and cost, not discrimination; (vi) public data only (no Rwandan/African clinical ECGs), so findings speak to cross-dataset sample efficiency, not local clinical performance; (vii) MIMIC-IV-ECG credentialed access can take days to weeks [4] credentialing starts at semester start, and if delayed past the M2 checkpoint (Table 3) the PTB-XL $\leftrightarrow$ Chapman-Shaoxing rotation is validated first, with MIMIC-IV-ECG pairs added once access is confirmed.

VIII. Project Milestones and Academic Timeline

The project follows the 04-990 Engineering Research Project structure: 36 cumulative units, including 6 already completed in the research methods course, plus 12 units in Fall 2026 and 18 in Spring 2027. Milestones, acceptance criteria, dates, and grading weights are given in the Appendix (Tables 3 and 4).

IX. Expected Project Deliverables

Reproducible 1D-ResNet baselines for all three source datasets (PTB-XL benchmarked against Table 2); unadapted cross-dataset evaluation with significance testing across all six pairs; Temperature Scaling, Linear Probing, and LoRA implementations; stabilization/improvement results across all five percentage levels with Monte Carlo uncertainty; $P_{\min}$ estimates; comparative figures/tables; a reproducible repository; and a mini-thesis with oral defense.

X. Research Significance and Conclusion

Rather than assuming cross-dataset ECG performance drops by a known amount, or that a large local dataset is always necessary, this study first quantifies whether and how much OOD performance changes, and whether it stays significantly above chance, before asking how it stabilizes and improves as target data increases building on external-validation concerns [1, 2] and positioned against existing PTB-XL benchmarking and transfer work [8–10]. It will not establish a universal clinical threshold, but a reproducible, percentage-based audit procedure – recording rotation, dataset pair, method, percentage, criterion, threshold, and uncertainty – repeatable across models and settings, consistent with external-validation literature holding that data needs depend on the model, outcome, metric, and intended use [1, 2].

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XI. Appendix

Assessment (100% per semester) and Milestones. Each row lists the deliverable together with its acceptance criterion; MIMIC-IV-ECG contingencies follow Section VII.

Table 3: Fall 2026: 12 Units

ID	Deliverable / Acceptance Criterion	Date	Wt.
M1	Extensive literature review; submit review report	Aug. 28	10%
M2	Baseline classifier trained on at least PTB-XL (macro-AUROC within Table 2 range) + preprocessing/label pipeline built to support all three datasets; MIMIC-IV-ECG PhysioNet credentialing initiated at semester start	Sep. 4	10%
M3	Unadapted OOD evaluation completed for ≥ 1 full source $\rightarrow$ target rotation (both targets), with bootstrap CIs and chance-level significance test; Temperature Scaling & Linear Probing implemented across all 5 percentage levels for that rotation, $\geq 10/30$ Monte Carlo reps as interim check	Oct. 9	30%
M4	Fall report + interim percentage-dependent stabilization/improvement curves + reproducibility package	Dec. 4	25%
M5	Progress presentation / mid-project defense	Dec. 14	25%

Table 4: Spring 2027: 18 Units

ID	Deliverable / Acceptance Criterion	Date	Wt.
M1	LoRA implemented (parameter count measured, not assumed); remaining two source→target rotations completed; full $3(\text{rotation}) \times 2(\text{target}) \times 3(\text{method}) \times 5(\%)$ Monte Carlo experiments underway	Jan. 18	20%
M2	Sensitivity/uncertainty analysis; $P_{\min}^{(\Delta)}$ and $\Delta M(P)$ (Eqs. 2–3) with CIs for every rotation/dataset/method combination; stabilization-curve fit diagnostics and descriptive source-gap $G(P)$ reported	Feb. 19	20%
M3	Final results/calibration tables across all conditions; mini-thesis manuscript (methods, results, discussion, limitations); submit final repository	Apr. 30	30%
M4	Final presentation and oral defense; graduation May 21, 2027	May 10	30%